

Population Growth

Can be modelled as a function of births & deaths

$$\Rightarrow f(b, d)$$

$P_t \rightarrow$ Population at time t .

$$\therefore \Delta P_{t+1} = P_{t+1} - P_t \quad \text{--- (1)}$$

$$\Delta P = B - D \quad \text{--- (1)}$$

Set $b =$ Per capita births
 $d =$ Per capita deaths

In (1),

$$\Delta P_t = bP_t - dP_t$$

$$\Delta P_t = (b-d)P_t$$

$$\Delta P_t = rP_t \quad ; \quad r = b-d \quad \text{--- (1)}$$

$$\Delta P(P) = rP$$

From (1),

$$r = \frac{\Delta P}{P}$$

Logistic $\Rightarrow$
 growth

$$\Delta P = rP(1-P)$$

$$\frac{dP}{dt} = rP(1-P)$$

$$\frac{dP}{dt} = rP - rP^2$$

$$\frac{dP}{rP - rP^2} = dt$$

$$\frac{1}{r} \int \frac{dP}{P(1-P)} = \int dt$$

on LHS by PFD,

$$\frac{1}{P(1-P)} = \frac{A}{P} + \frac{B}{1-P}$$

Multiply both
 sides by $P(1-P)$
 of LHS

$$1 = A(1-P) + B(P)$$

$$1 = A - AP + BP$$

$$1 = BP - AP + A$$

$$1 = (B-A)P + A$$

$$B-A = 0 \Rightarrow B = A$$

$$A = 1 \Rightarrow B = 1$$

$$\therefore \frac{1}{r} \int \frac{dP}{P} + \frac{dP}{1-P} = \int dt$$

$$\frac{1}{r} (\ln P + \ln(1-P)) = t + C$$

$$\frac{1}{r} \ln(P(1-P)) = t + C$$

$$\ln(P(1-P)^{1/2}) = t + C$$

$$t = \ln(P(1-P)^{1/2}) - C$$

$$t+C = \ln(P(1-P)^{1/2})$$

$$e^{r(t+C)} = P(1-P)$$

$$e^{rt} \cdot e^{rc} = P(1-P)$$

$$e^{r(t+C)} = P(1-P)$$

$$P(1-P) = e^{r(t+C)}$$

$$P(t) = \frac{e^{r(t+C)}}{1+e^{r(t+C)}}$$

Agent Based growth Model. (Forest growth)

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① Death rate $\rightarrow d \Rightarrow d_A$ or d_B

② Occupancy rate $\rightarrow o \Rightarrow o_A$ or o_B

For any two species A & B,

$$\Rightarrow A_{t+1} = A_t - d_A A_t + (d_A A_t + d_B B_t) o_A$$

$$\Rightarrow B_{t+1} = B_t - d_B B_t + (d_A A_t + d_B B_t) o_B$$

$$\begin{aligned} \text{Death term} &= d_A A_t \\ &= d_B B_t \end{aligned}$$

$$\begin{aligned} \text{Addition term} &= (d_A A_t + d_B B_t) o_A \\ &= (d_A A_t + d_B B_t) o_B \end{aligned}$$

2nd Session - 11:51 AM

In a logistic growth equation,

$$P_{t+1} = P_t + R \cdot P_t (1 - P_t)$$

Dividing both sides by Δt & $\Delta t \rightarrow 0$

$$\therefore \frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right) \quad ; \quad r = \frac{R}{\Delta t} \quad \& \quad K \rightarrow \text{carrying Capacity}$$

Continuous logistic growth ODE

Two species interaction

$$\textcircled{1} \frac{dP_1}{dt} = r_1 P_1 \left(\frac{K_1 - P_1 - \alpha_{12} P_2}{K_1} \right)$$

$$\textcircled{2} \frac{dP_2}{dt} = r_2 P_2 \left(\frac{K_2 - P_2 - \alpha_{21} P_1}{K_2} \right)$$

Various cases for P_n

1) If $P_1 = 0$, $P_2 = K_2$

2) If $P_2 = 0$, $P_1 = K_1$

$\Rightarrow$ This model supports coexistence,
when $P_1 \neq 0$ & $P_2 \neq 0$.

Doubts

$\rightarrow$ In later Vectors, what if the system of ODEs is converted to a system of PDEs?

$$P_1 = f(x, t)$$

$$P_2 = f(x, t)$$

$\rightarrow$ what would a second order ODE look like?
what can we infer from the curvature?

Predator-Prey Model

$$P_1 = \text{Prey}$$

$$P_2 = \text{Predator}$$

$$\begin{aligned} \frac{dP_1}{dt} &= r_1 P_1 - \alpha P_1 P_2 \\ \frac{dP_2}{dt} &= -r_2 P_2 + \beta P_1 P_2 \end{aligned}$$

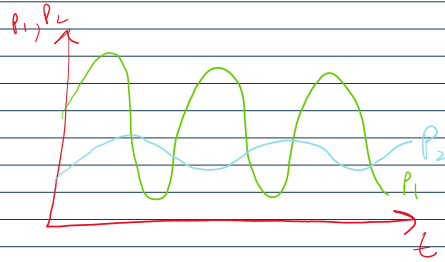
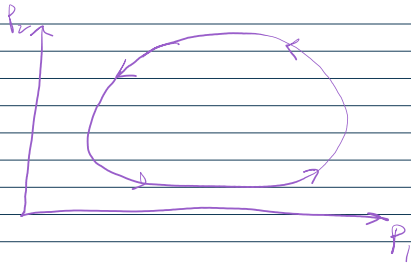
where, r_1 : per capita intrinsic growth rate of P_1

r_2 : per capita intrinsic death rate of P_2

α : Capture efficiency

β : Conversion efficiency

$\rightarrow$ Tells us what effect will consumption of one unit of prey have on the population growth.



Stochastic Modelling in Ecology

Goal $\rightarrow$ Beyond correlation between heights and weights ~

$$\sum P = 1$$

$$\frac{\Delta D}{\Delta t} = \frac{\ln(P_1) - \ln(P_2)}{t_2 - t_1}$$

$$\text{let } t_2 - t_1 = h$$

$$\therefore \frac{dD}{dt} = \lim_{h \rightarrow 0} \frac{\ln\left(\frac{P_1}{P_2}\right)}{h}$$



Strongly
Actually
same
 $\rightarrow$
12MB
BLIND
0.3.1415
 $\approx$ 1.77
= N LEE